

CASE STUDY: Bank Management System (BMS)

ABC Software Solutions Ltd. was contracted by **National Trust Bank (NTB)** to develop a **Bank Management System (BMS)** that would automate banking operations across all branches nationwide. The system was expected to improve customer service, increase operational efficiency, enhance transaction security, reduce manual processes, and support digital banking services. The project was planned for completion within **12 months** under a fixed budget while complying with all regulatory requirements issued by the Central Bank.

The proposed Bank Management System was designed to support the following core banking services:

- Customer Registration and Account Management
- Savings and Current Account Management
- Fixed Deposit Management
- Loan Application and Loan Management
- Fund Transfers (Internal and External)
- ATM and Debit/Credit Card Management
- Internet Banking
- Mobile Banking
- Cheque Processing
- Transaction Monitoring
- Financial Reporting
- Audit Trail and Compliance Reporting
- Integration with National Payment Systems
- Backup and Disaster Recovery

Because the system would process thousands of financial transactions every day, it was expected to achieve high standards of:

- Security
- Reliability
- Availability
- Performance
- Scalability
- Data Integrity
- Confidentiality
- Regulatory Compliance

During system development, the bank requested several additional features, including biometric customer authentication, real-time fraud detection, mobile wallet integration, AI-based credit scoring, and enhanced financial reporting. The development team accepted these changes without conducting a formal **Contract Review** or updating the project scope, budget, schedule, and risk assessment. Consequently, the project experienced scope creep, schedule delays, increased development costs, and frequent misunderstandings between the client and the software development team.

The project also suffered from weaknesses in software planning and quality management. Although a **Software Development Plan (SDP)** and a **Software Quality Assurance Plan (SQAP)** had been prepared, they were incomplete and not consistently followed. Different development teams used different coding standards, documentation formats, and development practices, making software integration difficult. Communication among project managers, developers, business analysts, quality assurance engineers, and stakeholders was inconsistent, resulting in incomplete requirements, duplicated work, delayed approvals, and unresolved technical issues.

Software testing was not properly planned or managed. Testing activities began late in the project, and the testing environment differed significantly from the production environment. Test data did not accurately represent real banking transactions, resulting

in unreliable test outcomes. Several important testing activities were either incomplete or skipped, including:

- Unit Testing
- Integration Testing
- System Testing
- Performance Testing
- Security Testing
- User Acceptance Testing (UAT)
- Regression Testing

Furthermore, many test cases were incomplete, defects were poorly documented, some defects were never assigned to developers, and several critical defects remained unresolved before deployment. Retesting and regression testing were inconsistently performed after bug fixes, causing previously resolved defects to reappear in subsequent software versions.

As the deployment date approached, the Quality Assurance team identified several critical issues:

- Incorrect processing of financial transactions
- Inconsistent customer account balances
- Unauthorized access to restricted banking functions
- Slow system response during peak transaction periods
- Incomplete audit logs
- Unverified backup and disaster recovery procedures
- Poorly documented test results
- Missing evidence that regulatory requirements had been satisfied
- Incomplete Test Summary Report

- High number of unresolved critical defects

Concerned about the potential financial, operational, legal, and reputational risks, the bank's executive management postponed the deployment of the Bank Management System. An independent **Software Quality Assurance (SQA) Team** was appointed to conduct a comprehensive quality assessment before approving the system for production.

The SQA team was assigned to perform the following tasks:

- Review the Contract Review process.
- Evaluate the Software Development Plan (SDP).
- Assess the Software Quality Assurance Plan (SQAP).
- Review the Software Testing Strategy.
- Evaluate the Software Testing Implementation process.
- Assess the Software Test Environment and Test Data.
- Review Defect Management practices.
- Analyze Software Quality Metrics and Key Performance Indicators (KPIs).
- Review the Test Summary Report and supporting documentation.
- Identify project risks and recommend mitigation strategies.
- Prepare a comprehensive Software Quality Improvement Plan.
- Determine whether the Bank Management System is ready for deployment or requires further quality improvements before release.

END